

MC001 · Pyramid Problems

Minecraft · Code Builder — From a one-line path to a full diamond structure with loops & variables.

For parents: these are the key English coding words your child meets in this course, week by week. You don't need to code — just read a word together and ask your child to explain it back in their own words. The tiny example shows what it looks like in real code.

W1

WORD	WHAT IT MEANS	TINY EXAMPLE
agent	the robot helper your code controls	<pre>agent.move(FORWARD, 1)</pre>
instruction	one command you give the agent	<pre>agent.place(DOWN)</pre>
run	make the code happen in the world	<pre>player.on_chat("start", go)</pre>

W2

WORD	WHAT IT MEANS	TINY EXAMPLE
loop	repeat code instead of retyping it	<pre>for i in range(8):</pre>
repeat	do the same block N times	<pre>for i in range(8): place()</pre>
iteration	one pass through the loop	<pre># each i = one block</pre>

W3

WORD	WHAT IT MEANS	TINY EXAMPLE
variable	a named box that holds a value	<pre>size = 5</pre>
value	the data stored in the variable	<pre>size = 5 # value is 5</pre>
reuse value	use one variable in many places	<pre>range(size)</pre>

W4

WORD	WHAT IT MEANS	TINY EXAMPLE
nested loop	a loop inside another loop	<pre>for r in range(3): for c in range(3):</pre>
outer / inner	outer = rows, inner = each block in a row	<pre>outer rows, inner cols</pre>
indentation	spaces that show what's inside a loop	<pre>for i in range(3): place()</pre>

W5

WORD	WHAT IT MEANS	TINY EXAMPLE
layer width	each pyramid layer is smaller	<code>range(size - level)</code>
step	the change between layers	<code>level += 1</code>

W6

WORD	WHAT IT MEANS	TINY EXAMPLE
range step	count up or down by an amount	<code>range(5, 0, -1)</code>
reverse	flip the order to invert the shape	<code>for i in range(5, 0, -1):</code>

W7

WORD	WHAT IT MEANS	TINY EXAMPLE
parameter	an input that changes the build size	<code>def pyramid(size):</code>
generalize	one function that builds any size	<code>pyramid(7)</code>

W8

WORD	WHAT IT MEANS	TINY EXAMPLE
combine	join two builds into one	<code>pyramid_up(); pyramid_down()</code>
mirror	a shape repeated upside-down	<code># top + flipped bottom</code>
plan	decide the steps before coding	<code># 1 base 2 layers 3 tip</code>